

This document describes the Application Programming Interface (API) for the supporting libraries used throughout the book *Computer Science: An Interdisciplinary Approach* [↗](#) by Robert Sedgewick and Kevin Wayne. The corresponding code is part of a package called `stdlib`.

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## 1 BinaryIn

|                          |                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------|
| BinaryIn()               | constructs a binary input stream from standard input                                                    |
| BinaryIn(InputStream in) | constructs a binary input stream from an input stream                                                   |
| BinaryIn(Socket socket)  | constructs a binary input stream from a socket                                                          |
| BinaryIn(URL url)        | constructs a binary input stream from an URL                                                            |
| BinaryIn(String name)    | constructs a binary input stream from a file or an URL with the given name                              |
| boolean exists()         | returns <b>true</b> if this binary input stream exists, and <b>false</b> otherwise                      |
| boolean isEmpty()        | returns <b>true</b> if this binary input stream is empty, and <b>false</b> otherwise                    |
| boolean readBoolean()    | reads and returns the next bit of data from this binary input stream as a boolean                       |
| char readChar()          | reads and returns the next 8 bits of data from this binary input stream as a char                       |
| char readChar(int r)     | reads and returns the next <b>r</b> bits of data from this binary input stream as an <b>r</b> -bit char |
| String readString()      | reads and returns the remaining bytes of data from this binary input stream as a string                 |
| short readShort()        | reads and returns the next 16 bits of data from this binary input stream as a short                     |
| int readInt()            | reads and returns the next 32 bits of data from this binary input stream as an int                      |
| int readInt(int r)       | reads and returns the next <b>r</b> bits of data from this binary input stream as an <b>r</b> -bit int  |
| long readLong()          | reads and returns the next 64 bits of data from this binary input stream as a long                      |
| double readDouble()      | reads and returns the next 64 bits of data from this binary input stream as a double                    |
| float readFloat()        | reads and returns the next 32 bits of data from this binary input stream as a float                     |
| byte readByte()          | reads and returns the next 8 bits of data from this binary input stream as a byte                       |

## 2 BinaryOut

|                             |                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------|
| BinaryOut()                 | constructs a binary output stream from standard output                                         |
| BinaryOut(OutputStream in)  | constructs a binary output stream from an output stream                                        |
| BinaryOut(String name)      | constructs a binary output stream from a file with the given name                              |
| BinaryOut(Socket socket)    | constructs a binary output stream from a socket                                                |
| void flush()                | flushes this binary output stream, padding 0s if number of bits written is not a multiple of 8 |
| void close()                | flushes and closes this binary output stream                                                   |
| void write(boolean x)       | writes the bit to this binary output stream                                                    |
| void write(byte x)          | writes the 8-bit byte to this binary output stream                                             |
| void write(int x)           | writes the 32-bit int to this binary output stream                                             |
| void write(int x, int r)    | writes the <b>r</b> -bit int to this binary output stream                                      |
| void write(double x)        | writes the 64-bit double to this binary output stream                                          |
| void write(long x)          | writes the 64-bit long to this binary output stream                                            |
| void write(float x)         | writes the 32-bit float to this binary output stream                                           |
| void write(short x)         | writes the 16-bit short to this binary output stream                                           |
| void write(char x)          | writes the 8-bit char to this binary output stream                                             |
| void write(char x, int r)   | writes the <b>r</b> -bit char to this binary output stream                                     |
| void write(String x)        | writes the string of 8-bit characters to this binary output stream                             |
| void write(String x, int r) | writes the string of <b>r</b> -bit characters to this binary output stream                     |

### 3 BinaryStdIn

|                                           |                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>static void close()</code>          | closes standard input and releases any associated resources                                   |
| <code>static boolean isEmpty()</code>     | returns <b>true</b> if standard input is empty, and <b>false</b> otherwise                    |
| <code>static boolean readBoolean()</code> | reads and returns the next bit of data from standard input as a boolean                       |
| <code>static char readChar()</code>       | reads and returns the next 8 bits of data from standard input as a char                       |
| <code>static char readChar(int r)</code>  | reads and returns the next <b>r</b> bits of data from standard input as an <b>r</b> -bit char |
| <code>static String readString()</code>   | reads and returns the remaining bytes of data from standard input as a string                 |
| <code>static short readShort()</code>     | reads and returns the next 16 bits of data from standard input as a short                     |
| <code>static int readInt()</code>         | reads and returns the next 32 bits of data from standard input as an int                      |
| <code>static int readInt(int r)</code>    | reads and returns the next <b>r</b> bits of data from standard input as an <b>r</b> -bit int  |
| <code>static long readLong()</code>       | reads and returns the next 64 bits of data from standard input as a long                      |
| <code>static double readDouble()</code>   | reads and returns the next 64 bits of data from standard input as a double                    |
| <code>static float readFloat()</code>     | reads and returns the next 32 bits of data from standard input as a float                     |
| <code>static byte readByte()</code>       | reads and returns the next 8 bits of data from standard input as a byte                       |

### 4 BinaryStdOut

|                                                 |                                                                                      |
|-------------------------------------------------|--------------------------------------------------------------------------------------|
| <code>static void flush()</code>                | flushes standard output, padding 0s if number of bits written is not a multiple of 8 |
| <code>static void close()</code>                | flushes and closes standard output                                                   |
| <code>static void write(boolean x)</code>       | writes the bit to standard output                                                    |
| <code>static void write(byte x)</code>          | writes the 8-bit byte to standard output                                             |
| <code>static void write(int x)</code>           | writes the 32-bit int to standard output                                             |
| <code>static void write(int x, int r)</code>    | writes the <b>r</b> -bit int to standard output                                      |
| <code>static void write(double x)</code>        | writes the 64-bit double to standard output                                          |
| <code>static void write(long x)</code>          | writes the 64-bit long to standard output                                            |
| <code>static void write(float x)</code>         | writes the 32-bit float to standard output                                           |
| <code>static void write(short x)</code>         | writes the 16-bit short to standard output                                           |
| <code>static void write(char x)</code>          | writes the 8-bit char to standard output                                             |
| <code>static void write(char x, int r)</code>   | writes the <b>r</b> -bit char to standard output                                     |
| <code>static void write(String x)</code>        | writes the string of 8-bit characters to standard output                             |
| <code>static void write(String x, int r)</code> | writes the string of <b>r</b> -bit characters to standard output                     |

### 5 In

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <code>In()</code>                | constructs an input stream from standard input                              |
| <code>In(Socket socket)</code>   | constructs an input stream from a socket                                    |
| <code>In(URL url)</code>         | constructs an input stream from an URL                                      |
| <code>In(String name)</code>     | constructs an input stream from a file with the given name                  |
| <code>In(Scanner scanner)</code> | constructs an input stream from a scanner                                   |
| <code>boolean exists()</code>    | returns <b>true</b> if this input stream exists, and <b>false</b> otherwise |

|                                        |                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>boolean isEmpty()</code>         | returns <code>true</code> if this input stream is empty, and <code>false</code> otherwise        |
| <code>boolean hasNextLine()</code>     | returns <code>true</code> if this input stream has a next line, and <code>false</code> otherwise |
| <code>boolean hasNextChar()</code>     | returns <code>true</code> if this input stream has a next char, and <code>false</code> otherwise |
| <code>String readLine()</code>         | reads and returns the next line from this input stream                                           |
| <code>char readChar()</code>           | reads and returns the next char from this input stream                                           |
| <code>String readAll()</code>          | reads and returns the remainder of this input stream as a string                                 |
| <code>String readString()</code>       | reads and returns the next token from this input stream as a string                              |
| <code>int readInt()</code>             | reads and returns the next int from this input stream                                            |
| <code>double readDouble()</code>       | reads and returns the next double from this input stream                                         |
| <code>float readFloat()</code>         | reads and returns the next float from this input stream                                          |
| <code>long readLong()</code>           | reads and returns the next long from this input stream                                           |
| <code>short readShort()</code>         | reads and returns the next short from this input stream                                          |
| <code>byte readByte()</code>           | reads and returns the next byte from this input stream                                           |
| <code>boolean readBoolean()</code>     | reads and returns the next boolean from this input stream                                        |
| <code>String[] readAllStrings()</code> | reads and returns all the remaining tokens from this input stream as an array of strings         |
| <code>String[] readAllLines()</code>   | reads and returns all the remaining lines from this input stream as an array of strings          |
| <code>int[] readAllInts()</code>       | reads and returns all the remaining tokens from this input stream as an array of ints            |
| <code>long[] readAllLongs()</code>     | reads and returns all the remaining tokens from this input stream as an array of longs           |
| <code>double[] readAllDoubles()</code> | reads and returns all the remaining tokens from this input stream as an array of doubles         |
| <code>void close()</code>              | closes this input stream                                                                         |

## 6 Out

|                                      |                                                             |
|--------------------------------------|-------------------------------------------------------------|
| <code>Out()</code>                   | constructs an output stream from standard output            |
| <code>Out(Socket socket)</code>      | constructs an output stream from a socket                   |
| <code>Out(String name)</code>        | constructs an output stream from a file with the given name |
| <code>void close()</code>            | closes this output stream                                   |
| <code>void println()</code>          | prints a newline to this output stream                      |
| <code>void println(Object x)</code>  | prints an object and a newline to this output stream        |
| <code>void println(boolean x)</code> | prints a boolean and a newline to this output stream        |
| <code>void println(char x)</code>    | prints a char and a newline to this output stream           |
| <code>void println(double x)</code>  | prints a double and a newline to this output stream         |
| <code>void println(float x)</code>   | prints a float and a newline to this output stream          |
| <code>void println(int x)</code>     | prints an int and a newline to this output stream           |
| <code>void println(long x)</code>    | prints a long and a newline to this output stream           |
| <code>void println(byte x)</code>    | prints a byte and a newline to this output stream           |
| <code>void print()</code>            | flushes this output stream                                  |
| <code>void print(Object x)</code>    | prints an object to this output stream                      |
| <code>void print(boolean x)</code>   | prints a boolean to this output stream                      |
| <code>void print(char x)</code>      | prints a char to this output stream                         |

|                                                                  |                                                                                                                 |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>void print(double x)</code>                                | prints a double to this output stream                                                                           |
| <code>void print(float x)</code>                                 | prints a float to this output stream                                                                            |
| <code>void print(int x)</code>                                   | prints an int to this output stream                                                                             |
| <code>void print(long x)</code>                                  | prints a long to this output stream                                                                             |
| <code>void print(byte x)</code>                                  | prints a byte to this output stream                                                                             |
| <code>void printf(String fmt, Object... args)</code>             | prints <code>args</code> to this output stream using format string <code>fmt</code>                             |
| <code>void printf(Locale loc, String fmt, Object... args)</code> | prints <code>args</code> to this output stream using locale <code>loc</code> and format string <code>fmt</code> |

## 7 Picture

|                                                     |                                                                                                                         |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>Picture(int w, int h)</code>                  | constructs a picture of width <code>w</code> and height <code>h</code>                                                  |
| <code>Picture(Picture pic)</code>                   | constructs a picture that is a deep copy of <code>pic</code>                                                            |
| <code>Picture(String name)</code>                   | constructs a picture from an image with the given name                                                                  |
| <code>JLabel getLabel()</code>                      | returns the current label of this picture for use in some other GUI                                                     |
| <code>void setOriginUpperLeft()</code>              | sets the origin to be the upper left pixel (default)                                                                    |
| <code>void setOriginLowerLeft()</code>              | sets the origin to be the lower left pixel                                                                              |
| <code>void show()</code>                            | displays this picture on the screen                                                                                     |
| <code>int height()</code>                           | returns the height of this picture                                                                                      |
| <code>int width()</code>                            | returns the width of this picture                                                                                       |
| <code>RGBColor get(int col, int row)</code>         | returns the color of pixel ( <code>col</code> , <code>row</code> ) as a <code>RGBColor</code> object                    |
| <code>void set(int col, int row, RGBColor c)</code> | sets the color of the pixel ( <code>col</code> , <code>row</code> ) to the given value (a <code>RGBColor</code> object) |
| <code>void save(String name)</code>                 | saves this picture to a file with the given name                                                                        |
| <code>void actionPerformed(ActionEvent e)</code>    | opens a save dialog box when the user selects “Save As” from the menu                                                   |

## 8 RGBColor

|                                                          |                                                                                                                 |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>RGBColor(int r, int g, int b)</code>               | constructs a color given its red, green, and blue components                                                    |
| <code>public int getRed()</code>                         | returns the red component of the color                                                                          |
| <code>public int getGreen()</code>                       | returns the green component of the color                                                                        |
| <code>public int getBlue()</code>                        | returns the blue component of the color                                                                         |
| <code>public double luminance()</code>                   | returns the luminance of the color                                                                              |
| <code>public RGBColor toGray()</code>                    | returns the grayscale equivalent of the color                                                                   |
| <code>public boolean isCompatible(RGBColor other)</code> | returns <code>true</code> if the color is compatible with <code>other</code> , and <code>false</code> otherwise |
| <code>public String toString()</code>                    | returns a string representation of the color                                                                    |

## 9 StdArrayIO

|                                                 |                                                                                                                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>static double[] readDouble1D()</code>     | reads an integer $n$ from standard input, and then reads $n$ doubles also from standard input and returns them as a 1D array of size $n$                  |
| <code>static void print(double[] a)</code>      | prints the size and elements of the 1D array <b>a</b> to standard output                                                                                  |
| <code>static double[][] readDouble2D()</code>   | reads integers $m$ and $n$ from standard input, and then reads $mn$ doubles also from standard input and returns them as a 2D array of size $m \times n$  |
| <code>static void print(double[][] a)</code>    | prints the size and elements of the 2D array <b>a</b> to standard output                                                                                  |
| <code>static int[] readInt1D()</code>           | reads an integer $n$ from standard input, and then reads $n$ ints also from standard input and returns them as a 1D array of size $n$                     |
| <code>static void print(int[] a)</code>         | prints the size and elements of the 1D array <b>a</b> to standard output                                                                                  |
| <code>static int[][] readInt2D()</code>         | reads integers $m$ and $n$ from standard input, and then reads $mn$ ints also from standard input and returns them as a 2D array of size $m \times n$     |
| <code>static void print(int[][] a)</code>       | prints the size and elements of the 2D array <b>a</b> to standard output                                                                                  |
| <code>static boolean[] readBoolean1D()</code>   | reads an integer $n$ from standard input, and then reads $n$ booleans also from standard input and returns them as a 1D array of size $n$                 |
| <code>static void print(boolean[] a)</code>     | prints the size and elements of the 1D array <b>a</b> to standard output                                                                                  |
| <code>static boolean[][] readBoolean2D()</code> | reads integers $m$ and $n$ from standard input, and then reads $mn$ booleans also from standard input and returns them as a 2D array of size $m \times n$ |
| <code>static void print(boolean[][] a)</code>   | prints the size and elements of the 2D array <b>a</b> to standard output                                                                                  |

## 10 StdAudio

|                                                              |                                                                                                    |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>static void close()</code>                             | closes standard audio                                                                              |
| <code>static void play(double sample)</code>                 | plays one sample (between -1.0 and +1.0) using standard audio                                      |
| <code>static void play(double[] samples)</code>              | plays an array of samples (between -1.0 and +1.0) using standard audio                             |
| <code>static double[] read(String name)</code>               | reads and returns samples from a file (.au or .wav format) with the given name                     |
| <code>static void save(String name, double[] samples)</code> | saves the samples as a file (.au or .wav format) with the given name                               |
| <code>static void play(String name)</code>                   | plays a file (.au, .mid, or .au format) with the given name in the background using standard audio |
| <code>static void loop(String name)</code>                   | loops a file (.au, .mid, or .au format) with the given name in the background using standard audio |

## 11 StdDraw

|                                                                                   |                                                                                                |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>static RGBColor BLACK</code>                                                | represents black                                                                               |
| <code>static RGBColor BLUE</code>                                                 | represents blue                                                                                |
| <code>static RGBColor CYAN</code>                                                 | represents cyan                                                                                |
| <code>static RGBColor DARK_GRAY</code>                                            | represents dark gray                                                                           |
| <code>static RGBColor GREEN</code>                                                | represents green                                                                               |
| <code>static RGBColor LIGHT_GRAY</code>                                           | represents light gray                                                                          |
| <code>static RGBColor MAGENTA</code>                                              | represents magenta                                                                             |
| <code>static RGBColor ORANGE</code>                                               | represents orange                                                                              |
| <code>static RGBColor PINK</code>                                                 | represents pink                                                                                |
| <code>static RGBColor RED</code>                                                  | represents red                                                                                 |
| <code>static RGBColor WHITE</code>                                                | represents white                                                                               |
| <code>static RGBColor YELLOW</code>                                               | represents yellow                                                                              |
| <code>static void setCanvasSize()</code>                                          | sets the width and height of the canvas to 512 pixels                                          |
| <code>static void setCanvasSize(int w, int h)</code>                              | sets the width and height of the canvas to <code>w</code> and <code>h</code> pixels            |
| <code>static void setXscale()</code>                                              | sets the <i>x</i> -scale of the canvas to the default <code>([0, 1])</code>                    |
| <code>static void setYscale()</code>                                              | sets the <i>y</i> -scale of the canvas to the default <code>([0, 1])</code>                    |
| <code>static void setScale()</code>                                               | sets the <i>x</i> - and <i>y</i> -scale of the canvas to the default <code>([0, 1])</code>     |
| <code>static void setXscale(double min, double max)</code>                        | sets the <i>x</i> -scale of the canvas to <code>[min, max]</code>                              |
| <code>static void setYscale(double min, double max)</code>                        | sets the <i>y</i> -scale of the canvas to <code>[min, max]</code>                              |
| <code>static void setScale(double min, double max)</code>                         | sets the <i>x</i> - and <i>y</i> -scale of the canvas to <code>[min, max]</code>               |
| <code>static void clear()</code>                                                  | clears the canvas to the default (white) color                                                 |
| <code>static void clear(RGBColor c)</code>                                        | clears the canvas to the color <code>c</code>                                                  |
| <code>static double getPenRadius()</code>                                         | returns the current pen radius of the canvas                                                   |
| <code>static void setPenRadius()</code>                                           | sets the pen radius of the canvas to the default (0.002)                                       |
| <code>static void setPenRadius(double r)</code>                                   | sets the pen radius of the canvas to <code>r</code>                                            |
| <code>static RGBColor getPenColor()</code>                                        | returns the current pen color of the canvas                                                    |
| <code>static void setPenColor()</code>                                            | sets the pen color of the canvas to the default (black) color                                  |
| <code>static void setPenColor(RGBColor c)</code>                                  | sets the pen color of the canvas to the color <code>c</code>                                   |
| <code>static void setPenColor(int r, int g, int b)</code>                         | sets the pen color of the canvas to the <code>(r, g, b)</code> color                           |
| <code>static void getFont()</code>                                                | returns the current font for the canvas                                                        |
| <code>static void setFont()</code>                                                | sets the font for the canvas to the default (sans serif, 16 point) font                        |
| <code>static void setFont(Font f)</code>                                          | sets the font for the canvas to the font <code>f</code>                                        |
| <code>static void drawLine(double x0, double y0,<br/>double x1, double y1)</code> | draws a line on the canvas from <code>(x0, y0)</code> to <code>(x1, y1)</code>                 |
| <code>static void draw(double x, double y)</code>                                 | draws one pixel on the canvas at <code>(x, y)</code>                                           |
| <code>static void point(double x, double y)</code>                                | draws a point on the canvas at <code>(x, y)</code>                                             |
| <code>static void circle(double x, double y, double r)</code>                     | draws a circle on the canvas of radius <code>r</code> , centered at <code>(x, y)</code>        |
| <code>static void filledCircle(double x, double y,<br/>double r)</code>           | draws a filled circle on the canvas of radius <code>r</code> , centered at <code>(x, y)</code> |

|                                                                                                        |                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>static void ellipse(double x, double y, double maj, double min)</code>                           | draws an ellipse on the canvas with semimajor and semiminor axes <code>maj</code> and <code>min</code> , centered at <code>(x, y)</code>       |
| <code>static void filledEllipse(double x, double y, double maj, double min)</code>                     | draws a filled ellipse on the canvas with semimajor and semiminor axes <code>maj</code> and <code>min</code> , centered at <code>(x, y)</code> |
| <code>static void arc(double x, double y, double r, double a1, double a2)</code>                       | draws an arc on the canvas with radius <code>r</code> , centered at <code>(x, y)</code> , from <code>a1</code> to <code>a2</code> (in degrees) |
| <code>static void square(double x, double y, double l)</code>                                          | draws a square on the canvas of side length <code>2l</code> , centered at <code>(x, y)</code>                                                  |
| <code>static void filledSquare(double x, double y, double l)</code>                                    | draws a filled square on the canvas of side length <code>2l</code> , centered at <code>(x, y)</code>                                           |
| <code>static void rectangle(double x, double y, double w, double h)</code>                             | draws a rectangle on the canvas of width <code>2w</code> and height <code>2h</code> , centered at <code>(x, y)</code>                          |
| <code>static void filledRectangle(double x, double y, double w, double h)</code>                       | draws a filled rectangle on the canvas of width <code>2w</code> and height <code>2h</code> , centered at <code>(x, y)</code>                   |
| <code>s</code> on the canvas, centered at <code>(x, y)</code> , rotated by <code>a</code> (in degrees) |                                                                                                                                                |
| <code>static void textLeft(double x, double y, String s)</code>                                        | draws the text <code>s</code> on the canvas, left-aligned at <code>(x, y)</code>                                                               |
| <code>static void textRight(double x, double y, String s)</code>                                       | draws the text <code>s</code> on the canvas, right-aligned at <code>(x, y)</code>                                                              |
| <code>static void pause(int t)</code>                                                                  | pauses the canvas for <code>t</code> milliseconds                                                                                              |
| <code>static void show()</code>                                                                        | copies offscreen buffer to onscreen buffer on the canvas                                                                                       |
| <code>static void enableDoubleBuffering()</code>                                                       | enables double buffering on the canvas                                                                                                         |
| <code>static void disableDoubleBuffering()</code>                                                      | disables double buffering on the canvas                                                                                                        |
| <code>static void save(String name)</code>                                                             | saves the drawing on the canvas to a file with the given name                                                                                  |
| <code>static boolean isMousePressed()</code>                                                           | returns <code>true</code> if the mouse is being pressed on the canvas, and <code>false</code> otherwise                                        |
| <code>static double mouseX()</code>                                                                    | returns the <i>x</i> -coordinate of the mouse pointer on the canvas                                                                            |
| <code>static double mouseY()</code>                                                                    | returns the <i>y</i> -coordinate of the mouse pointer on the canvas                                                                            |
| <code>static boolean hasNextKeyTyped()</code>                                                          | returns <code>true</code> if the user has typed a key on the canvas, and <code>false</code> otherwise                                          |
| <code>static char nextKeyTyped()</code>                                                                | returns the next key typed by the user on the canvas                                                                                           |
| <code>static boolean isKeyPressed(int c)</code>                                                        | returns <code>true</code> if the keycode <code>c</code> is being pressed on the canvas, and <code>false</code> otherwise                       |

## 12 StdIn

|                                           |                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>static boolean isEmpty()</code>     | returns <code>true</code> if standard input is empty, and <code>false</code> otherwise        |
| <code>static boolean hasNextLine()</code> | returns <code>true</code> if standard input has a next line, and <code>false</code> otherwise |
| <code>static boolean hasNextChar()</code> | returns <code>true</code> if standard input has a next char, and <code>false</code> otherwise |
| <code>static String readLine()</code>     | reads and returns the next line from standard input                                           |
| <code>char readChar()</code>              | reads and returns the next char from this standard input                                      |
| <code>static String readAll()</code>      | reads and returns the remainder of standard input as a string                                 |



|                                               |                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------------|
| <code>static String readString()</code>       | reads and returns the next token from standard input as a string                      |
| <code>static int readInt()</code>             | reads and returns the next int from standard input                                    |
| <code>static double readDouble()</code>       | reads and returns the next double from standard input                                 |
| <code>static float readFloat()</code>         | reads and returns the next float from standard input                                  |
| <code>static long readLong()</code>           | reads and returns the next long from standard input                                   |
| <code>static short readShort()</code>         | reads and returns the next short from standard input                                  |
| <code>static byte readByte()</code>           | reads and returns the next byte from standard input                                   |
| <code>static boolean readBoolean()</code>     | reads and returns the next boolean from standard input                                |
| <code>static String[] readAllStrings()</code> | reads and returns all the remaining tokens from standard input as an array of strings |
| <code>static String[] readAllLines()</code>   | reads and returns all the remaining lines from standard input as an array of strings  |
| <code>static int[] readAllInts()</code>       | reads and returns all the remaining tokens from standard input as an array of ints    |
| <code>static long[] readAllLongs()</code>     | reads and returns all the remaining tokens from standard input as an array of longs   |
| <code>static double[] readAllDoubles()</code> | reads and returns all the remaining tokens from standard input as an array of doubles |
| <code>static void resync()</code>             | reinitializes the scanner underlying standard input                                   |

## 13 StdOut

|                                             |                                                   |
|---------------------------------------------|---------------------------------------------------|
| <code>static void println()</code>          | prints a newline to standard output               |
| <code>static void println(Object x)</code>  | prints an object and a newline to standard output |
| <code>static void println(boolean x)</code> | prints a boolean and a newline to standard output |
| <code>static void println(char x)</code>    | prints a char and a newline to standard output    |
| <code>static void println(double x)</code>  | prints a double and a newline to standard output  |
| <code>static void println(float x)</code>   | prints a float and a newline to standard output   |
| <code>static void println(int x)</code>     | prints an int and a newline to standard output    |
| <code>static void println(long x)</code>    | prints a long and a newline to standard output    |
| <code>static void println(short x)</code>   | prints a short and a newline to standard output   |
| <code>static void println(byte x)</code>    | prints a byte and a newline to standard output    |
| <code>static void print()</code>            | flushes standard output                           |
| <code>static void print(Object x)</code>    | prints an object to standard output               |
| <code>static void print(boolean x)</code>   | prints a boolean to standard output               |
| <code>static void print(char x)</code>      | prints a char to standard output                  |
| <code>static void print(double x)</code>    | prints a double to standard output                |
| <code>static void print(float x)</code>     | prints a float to standard output                 |
| <code>static void print(int x)</code>       | prints an int to standard output                  |
| <code>static void print(long x)</code>      | prints a long to standard output                  |
| <code>static void print(short x)</code>     | prints a short to standard output                 |
| <code>static void print(byte x)</code>      | prints a byte to standard output                  |

|                                                                         |                                                                                            |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <code>static void printf(String fmt, Object... args)</code>             | prints <b>args</b> to standard output using format string <b>fmt</b>                       |
| <code>static void printf(Locale loc, String fmt, Object... args)</code> | prints <b>args</b> to standard output using locale <b>loc</b> and format string <b>fmt</b> |
| <code>static void resync()</code>                                       | reinitializes the writer underlying standard output                                        |

## 14 StdRandom

|                                                              |                                                                                                       |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>static seed(long s)</code>                             | sets the seed of the random number generator to <b>s</b>                                              |
| <code>static long getSeed()</code>                           | returns the seed of the random number generator                                                       |
| <code>static double uniform()</code>                         | returns a double chosen uniformly at random from the interval <b>[0, 1)</b>                           |
| <code>static int uniform(int n)</code>                       | returns an integer chosen uniformly at random from the interval <b>[0, n)</b>                         |
| <code>static long uniform(long n)</code>                     | returns a long chosen uniformly at random from the interval <b>[0, n)</b>                             |
| <code>static int uniform(int a, int b)</code>                | returns an integer chosen uniformly at random from the interval <b>[a, b)</b>                         |
| <code>static double uniform(double a, double b)</code>       | returns a double chosen uniformly at random from the interval <b>[a, b)</b>                           |
| <code>static boolean bernoulli(double p)</code>              | returns <b>true</b> with probability <b>p</b> and <b>false</b> with probability <b>1 - p</b>          |
| <code>static boolean bernoulli()</code>                      | returns <b>true</b> with probability 0.5 and <b>false</b> with probability 0.5                        |
| <code>static double gaussian()</code>                        | returns a double from the standard Gaussian distribution                                              |
| <code>static double gaussian(double mu, double sigma)</code> | returns a double from a Gaussian distribution with mean <b>mu</b> and standard deviation <b>sigma</b> |
| <code>static int geometric(double p)</code>                  | returns an integer from a geometric distribution with success probability <b>p</b>                    |
| <code>static int poisson(double lambda)</code>               | returns an integer from a Poisson distribution with mean <b>lambda</b>                                |
| <code>static double pareto()</code>                          | returns a double from the standard Pareto distribution                                                |
| <code>static double pareto(double alpha)</code>              | returns a double from a Pareto distribution with shape parameter <b>alpha</b>                         |
| <code>static double cauchy()</code>                          | returns a double from the Cauchy distribution                                                         |
| <code>static int discrete(double[] probabilities)</code>     | returns an integer <b>i</b> with probability <b>probabilities[i]</b>                                  |
| <code>static int discrete(int[] frequencies)</code>          | returns an integer <b>i</b> with probability <b>frequencies[i]</b>                                    |
| <code>static double exp(double lambda)</code>                | returns a double from an exponential distribution with rate <b>lambda</b>                             |
| <code>static void shuffle(Object[] a)</code>                 | shuffles the array <b>a</b>                                                                           |
| <code>static void shuffle(double[] a)</code>                 | shuffles the array <b>a</b>                                                                           |
| <code>static void shuffle(int[] a)</code>                    | shuffles the array <b>a</b>                                                                           |
| <code>static void shuffle(Object[] a, int lo, int hi)</code> | shuffles the subarray <b>a[lo, ..., hi)</b>                                                           |

|                                                              |                                                                                            |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <code>static void shuffle(double[] a, int lo, int hi)</code> | shuffles the subarray <code>a[lo, ..., hi]</code>                                          |
| <code>static void shuffle(int[] a, int lo, int hi)</code>    | shuffles the subarray <code>a[lo, ..., hi]</code>                                          |
| <code>static int[] permutation(int n)</code>                 | returns a uniformly random permutation of <code>n</code> elements                          |
| <code>static int[] permutation(int n, int k)</code>          | returns a uniformly random permutation of <code>k</code> (out of <code>n</code> ) elements |

## 15 StdStats

|                                                                |                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>static double max(double[] a)</code>                     | returns the maximum value in the array <code>a</code>                                 |
| <code>static double max(double[] a, int lo, int hi)</code>     | returns the maximum value in the subarray <code>a[lo, ..., hi]</code>                 |
| <code>static int max(int[] a)</code>                           | returns the maximum value in the array <code>a</code>                                 |
| <code>static double min(double[] a)</code>                     | returns the minimum value in the array <code>a</code>                                 |
| <code>static double min(double[] a, int lo, int hi)</code>     | returns the minimum value in the subarray <code>a[lo, ..., hi]</code>                 |
| <code>static int min(int[] a)</code>                           | returns the minimum value in the array <code>a</code>                                 |
| <code>static double mean(double[] a)</code>                    | returns the average value in the array <code>a</code>                                 |
| <code>static double mean(double[] a, int lo, int hi)</code>    | returns the average value in the subarray <code>a[lo, ..., hi]</code>                 |
| <code>static double mean(int[] a)</code>                       | returns the average value in the array <code>a</code>                                 |
| <code>static double var(double[] a)</code>                     | returns the sample variance in the array <code>a</code>                               |
| <code>static double var(double[] a, int lo, int hi)</code>     | returns the sample variance in the subarray <code>a[lo, ..., hi]</code>               |
| <code>static double var(int[] a)</code>                        | returns the sample variance in the array <code>a</code>                               |
| <code>static double varp(double[] a)</code>                    | returns the population variance in the array <code>a</code>                           |
| <code>static double varp(double[] a, int lo, int hi)</code>    | returns the population variance in the subarray <code>a[lo, ..., hi]</code>           |
| <code>static double stddev(double[] a)</code>                  | returns the sample standard deviation in the array <code>a</code>                     |
| <code>static double stddev(double[] a, int lo, int hi)</code>  | returns the sample standard deviation in the subarray <code>a[lo, ..., hi]</code>     |
| <code>static double stddev(int[] a)</code>                     | returns the sample standard deviation in the array <code>a</code>                     |
| <code>static double stddevp(double[] a)</code>                 | returns the population standard deviation in the array <code>a</code>                 |
| <code>static double stddevp(double[] a, int lo, int hi)</code> | returns the population standard deviation in the subarray <code>a[lo, ..., hi]</code> |
| <code>static double sum(double[] a)</code>                     | returns the sum of all values in the array <code>a</code>                             |
| <code>static double sum(double[] a, int lo, int hi)</code>     | returns the sum of all values in the subarray <code>a[lo, ..., hi]</code>             |
| <code>static int sum(int[] a)</code>                           | returns the sum of all values in the array <code>a</code>                             |
| <code>static void plotPoints(double[] a)</code>                | plots the values in the array <code>a</code> as points                                |
| <code>static void plotLines(double[] a)</code>                 | plots the values in the array <code>a</code> as line end-points                       |
| <code>static void plotBars(double[] a)</code>                  | plots the values in the array <code>a</code> as bars                                  |